

The effect of parental death on investments in children's education: Evidence from Peru

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October 2025

Abstract

This study investigates how the death of a parent affects children's intra-household resource allocation and human capital development in Peru. Utilizing longitudinal data from the younger cohort of the Young Lives study, we implement a staggered Difference-in-Differences framework to analyze seven outcomes: educational expenditures, school enrollment, daily attendance, home-study hours, domestic chores, and both remunerated and non-remunerated child labor. The results show that parental loss triggers persistent negative effects concentrated strictly on education, significantly reducing school enrollment and daily attendance, alongside a negative but non-significant downward trend in educational spending. Regarding the underlying mechanisms, we find that while the economic wealth channel plays a relevant role, it is not the sole driver. Shifting household expectations regarding the returns to education also serve as a key mechanism, as evidenced by a more pronounced enrollment drop among children with lower pre-shock academic performance.

Keywords: Parental death, Child education, Human capital, Family shocks

1 Introduction

The death of one of the parents constitutes one of the most significant adversities that a child or adolescent may experience throughout their development. Both in the social imaginary and in literature, this experience is usually addressed exclusively from a psychological perspective, focusing mainly on the emotional aftermath it leaves behind. However, empirical evidence indicates that its effects can be multiple and profound, affecting various dimensions of development, whose intensity depends, to a large extent, on the moment in which the loss occurs.

When the loss of a father or mother occurs during childhood or adolescence, its long-lasting consequences extend across multiple life domains, as documented by empirical literature. For example, a study conducted in Finland found that losing a parent before the age of 21 is associated with a higher probability of being diagnosed with a mental disorder, which carries consequences in adulthood, such as lower performance in the labor market, partly due to increased hospitalizations linked to these conditions (Böckerman et al., 2023).

Regarding educational development, various studies have documented that orphaned children are less likely to attend school, have higher school dropout rates, and significantly lower academic performance. In this regard, Campbell et al. (2010), based on the analysis of data from 11 countries in eastern and southern Africa, found that orphaned children were less likely to attend school and exhibited higher dropout rates. Although other factors such as poverty influence these outcomes, orphanhood remains an important determinant of educational lag.

While the aforementioned studies highlight direct impacts on psychological and educational dimensions, parental death also entails structural consequences for the accumulation of human capital and how the family unit reorganizes. This study aims to fill this specific gap in the literature. In this context, the present research seeks to answer the following question: What are the effects of parental death on children's investment in human capital? As a starting point, we hypothesize that the death of a parent has a negative effect

on educational investment. The main proposed transmission channel is economic in nature, derived from the loss of income generated by the deceased adult.

Despite the relevance of the topic, the literature on the effects of parental death in Latin America, and particularly in Peru, remains scarce, as will be shown in the literature review. This lack of evidence is especially concerning in contexts with fragmented social protection systems, where families often face major difficulties in absorbing severe shocks such as the loss of a father or mother. In Peru, for example, it is estimated that there are around 240,000 orphaned children (Andina, 2024), of whom only 100,000 receive some type of support from the State, mainly through financial assistance (Diario Oficial El Peruano, 2025).

Given this reality, it is essential to develop studies that identify the causal effects of the mechanisms through which orphanhood affects key variables such as educational outcomes. These studies would not only fill a gap in the academic literature, but could also inform public entities such as the Ministry of Development and Social Inclusion (MIDIS), whose mandate is oriented toward designing and implementing protection policies for vulnerable populations based on empirical evidence.

In terms of structure, this research is organized as follows. Section 2 reviews the relevant literature, covering theoretical approaches and empirical evidence. Section 3 presents the analytical framework. Section 4 outlines the empirical strategy, detailing the identification assumptions and the econometric model specification. Section 5 presents the data and variables, describing the sample selection process, variable definitions, and the baseline balance test. Section 6 discusses the empirical results, whereas Section 7 presents the robustness checks. Finally, Section 8 details the study's limitations and Section 9 offers the conclusions and recommendations for future research.

2 Literature Review

2.1 Theoretical Evidence

The effects of parental death on children have been addressed from various perspectives in the economic literature, giving rise to a wide range of theoretical approaches. These studies tend to differ in the assumptions they adopt regarding parents' preferences, including the form of their utility function, the discount rate applied to the welfare of future generations, the existence of perfect and complete capital markets, as well as the mechanisms through which parents invest in their children.

First, Becker and Tomes (1979), pioneers in the microeconomic analysis of family decisions, propose that households derive utility both from their own consumption and from the level of economic well-being their children will achieve in the future, which can be influenced through investments in their human capital. Within this framework, the optimal level of investment is determined at the point where the marginal return on educational investment equals its marginal cost. A key assumption of this model is the existence of perfect capital markets, which allows households to borrow without restrictions using the expected future income trajectory of their children as collateral. Under this assumption, negative shocks to current income, such as the death of one parent, do not constrain the possibility of reaching the optimal level of educational investment.

On the other hand, Michel et al. (2004) develop a model in which children do not inherit their parents' debts, meaning that parents cannot borrow using their descendants' future income as collateral. Within this framework, parents allocate resources to their children up to the point where the marginal utility of such allocation equals the marginal utility of their own consumption. Unlike the model with perfect capital markets, this approach introduces credit constraints that make children more vulnerable to negative shocks in household income, such as the death of one parent, because these shocks directly reduce the resources available to invest in them.

A third line of analysis considers contexts in which, even when households can borrow using their children’s future income as collateral, investment in children remains sensitive to income shocks due to the existence of imperfect capital markets and liquidity constraints. Becker and Tomes (1986) show that, under these conditions, families face difficulties financing the optimal level of investment in human capital. This effect could be mitigated if complete markets existed that allowed households to access insurance mechanisms, which would guarantee an efficient allocation of resources even in the face of adverse shocks (Townsend, 1995).

Returning to the model of Michel et al. (2004), another relevant contribution lies in the treatment of the discount factor applied to the utility of future generations and in the interpretation the authors assign to it. In their framework, individuals derive utility both from their own consumption and from the welfare of their children; however, since children also value the welfare of their own descendants, the utility function acquires a dynamic and recursive structure. This allows the problem to be reformulated such that the utility of each generation depends not only on its own welfare but also on that of future generations.

This recursive structure implies that the welfare of each future generation is valued differently. The authors introduce an intergenerational discount factor $\lambda \in (0, 1)$, which weights the utility of each descendant generation according to its temporal distance. Thus, the utility of children is discounted at a rate λ , that of grandchildren at λ^2 , and so on, progressively reducing the value assigned to more distant generations. This formulation is consistent with Hamilton’s Rule, according to which an individual will be willing to incur personal costs if doing so benefits another individual with whom they share a significant proportion of genetic material (Hamilton, 1964).

Within this framework, the model suggests that the more distant the relationship between the caregiver and the child, whether in temporal or biological terms, the lower the willingness to invest in the child’s welfare. This point becomes especially relevant in contexts of orphanhood, where the death of one parent may leave the child under the care of an agent

with a more distant relationship, potentially reducing investment in the child’s educational development and, consequently, in human capital accumulation.

Bolt et al. (2023) expand the theoretical analysis by developing an intertemporal life-cycle model that incorporates different stages—childhood, young adulthood, parenthood, and older adulthood—together with the resources available and the relevant decisions at each stage. In particular, their focus on the parenthood stage introduces an innovative classification of the resources parents allocate to their children, distinguishing them into three categories: direct monetary transfers, investment in education, and time devoted to human capital development. This more detailed representation of parental decisions makes it possible to identify that time investment and education have a greater impact on the child’s future welfare than purely monetary transfers.

The authors also identify a complementary relationship between parental involvement and the returns to educational investment. In contexts with imperfect capital markets, families facing liquidity constraints may be forced to reduce the time they devote to the child’s development because they lack the necessary resources (or the borrowing capacity) to simultaneously sustain parental involvement and the desired educational investment. This situation not only decreases investment in human capital but also reduces its returns, generating a vicious circle in which each component amplifies the losses of the other. Thus, negative shocks to household resources, such as those resulting from the death of one parent, simultaneously affect both parental involvement in the child’s development and educational investment, with long-lasting consequences for the child’s future income trajectory.

In summary, the theoretical models reviewed provide different conceptual frameworks for understanding how the loss of one parent may affect the formation of children’s human capital. However, the predictions depend heavily on the assumptions adopted, such as the existence of perfect capital markets, family structure, or intergenerational preferences, which generates diverse and, in some cases, ambiguous results. This heterogeneity limits the possibility of establishing generalizable conclusions based solely on theory. Therefore, it is

necessary to turn to the empirical literature to obtain a more precise understanding of the effects observed in real-world contexts.

2.2 Empirical Evidence

Various studies have documented how the death of a parent negatively impacts different dimensions of children’s human capital development, with education being one of the most extensively explored. Case, Anne and Ardington, Cally (2006), using two-round panel data from South Africa and fixed-effects models, find that children orphaned by their mother are less likely to be enrolled in school and, on average, complete fewer years of schooling. In addition, among children who do attend school, orphans receive lower educational investment in areas such as transportation, school supplies, and uniforms compared to their non-orphan peers. These patterns are not observed in the case of paternal loss. In contrast to these findings, Ainsworth et al. (2005), in a study of children in Tanzania, do not find significant effects of orphan status or the death of an adult household member on school enrollment. However, they do identify that orphans experience delays in starting primary education, as well as reductions in school attendance hours in the months preceding the death, although these recover once the death occurs.

For their part, Case et al. (2004) study the effect of orphanhood on school enrollment in ten countries in sub-Saharan Africa, with special attention to the underlying mechanisms. Initially, they find that orphaned children are less likely to be enrolled compared to non-orphans. Since previous estimates show that orphans, on average, live in poorer households, the authors evaluate whether the educational disadvantage can be explained mainly by household wealth. However, after controlling for household wealth, enrollment gaps persist, even among children living in mixed households (households hosting both orphans and non-orphans). This suggests that although household economic status influences school attendance, it is neither the only nor the most determining factor.

Subsequently, the authors identify a second explanatory mechanism: family arrangement.

In particular, they find that orphans living with household heads who are distant relatives have lower enrollment rates than those living with parents or close relatives. This finding is reinforced by Novella (2018), who, through an analysis controlling for household wealth, household characteristics, and community-level fixed effects, also finds that orphans are less likely to attend school, an effect that worsens in households where orphans live alongside non-orphan children. According to the author, the main factor associated with this difference is discrimination within the household, explained by the absence of biological closeness between the orphan and the household head. Finally, Case et al. (2004) raise the possibility that orphans receive lower educational investment because the expected returns to their education may be low as a consequence of adverse experiences associated with parental loss. However, this hypothesis is not verified due to limitations in the availability of data on children's abilities.

Another relevant study in the literature is that of Evans and Miguel (2007), who analyze the impact of parental death on school attendance using a five-year panel in Kenya. Their results show that school participation among orphans begins to decline two years before the parent's death, drops significantly in the year of death, and remains below that of non-orphan children for at least three years afterward. Although the death of either parent has negative effects on attendance, the impact of maternal death is approximately twice as large as that of paternal death. While these findings are consistent with previous evidence, the study's most novel contribution is that children with low academic performance at the beginning of the observation period experience a significantly greater decline in school attendance after parental loss.

Himaz (2013) analyzes whether the impact of parental death on children's educational and psychosocial outcomes varies depending on whether it occurs during childhood or adolescence, using panel data from the Young Lives study in Ethiopia. The results show that maternal death during childhood has a significant negative effect on school enrollment, whereas paternal death does not have a statistically relevant impact. According to the author, this

may be explained by the fact that, in the sample, many children whose mothers die are taken in by caregivers who are not close relatives, reinforcing the household structure mechanism. On the other hand, paternal death during childhood negatively affects the child's sense of optimism about the future. Although these effects are significant, they do not persist into adolescence. However, children who experienced paternal loss in childhood show greater participation in paid work and self-employment, which could suggest an economic channel. In contrast, during adolescence, the effects reverse, as maternal death does not affect school enrollment, whereas paternal death has a negative impact on enrollment, mathematics performance, and adolescents' sense of agency. Nevertheless, the channel through which these effects operate remains unclear, since no increase in employment or significant changes in household structure are observed. The author suggests that these impacts may be linked to psychosocial factors, such as lower perceived self-efficacy, although this mechanism is not explored in depth.

In summary, the reviewed literature provides consistent evidence regarding the negative effects of parental loss on children's education; however, important gaps remain that justify the development of this study. On the one hand, the evidence is concentrated in African countries, and other contexts have not been explored. On the other hand, many studies use methodological strategies that do not adequately account for variation in the timing of death across individuals, which may bias the estimation of the average effect. This study addresses these limitations by employing a methodology that corrects these issues and by applying it to the Peruvian context.

3 Analytical Framework

The loss of a parental figure during childhood constitutes a severe and disruptive event that profoundly alters household dynamics. In the economic literature, this type of occurrence is classified as an idiosyncratic shock (Dercon, 2002) because it occurs unexpectedly and affects

only one household, without being linked to aggregate macroeconomic conditions.

One of the areas most vulnerable to this type of shock is investment in children’s education. From the human capital theory perspective, such investment—which includes decisions such as school continuation, regular attendance, and time devoted to studying—is crucial because it enables the accumulation of skills and knowledge that increase future productivity and income-generating capacity throughout the life cycle (Bolt et al., 2023).

These decisions, made during early stages such as childhood and adolescence, are particularly sensitive due to their cumulative nature: disruptions during this phase can create setbacks that are difficult to reverse later on (Cunha and Heckman, 2007). The death of a parent often has immediate effects, such as a sudden reduction in household income, the reorganization of caregiving roles, and changes in the household’s strategic decisions, including those related to the human capital development of children.

3.1 Transmission Mechanisms

3.2 Household Resources

In general, parents provide the economic resources that sustain households. The death of one parent implies a direct reduction in available income, which can negatively affect multiple aspects of family life, including investment in education. This effect is exacerbated when the household faces liquidity constraints, as the income shock may prevent financing an optimal level of educational investment for children.

The decline in educational investment may primarily translate into a lower probability of school enrollment, especially if parents cannot afford tuition costs. However, even when enrollment does not involve direct expenses and children attend school, education may still be affected by other associated costs. For example, the inability to cover transportation expenses may reduce a child’s regular attendance, while a diminished capacity to meet basic needs such as nutrition or healthcare may lead to deteriorating health and increased school

absenteeism. Likewise, reduced spending on school supplies and educational materials limits the conditions necessary for learning. All these factors affect the quality of the educational process, particularly by reducing the accumulation of knowledge and skills, which in the long run compromises and diminishes the child’s educational and labor-market opportunities in later stages of life.

3.3 Family Structure

The death of a parent may also bring about significant changes in household structure. In some cases, the surviving parent forms a new partnership; in others, the child is sent to live with maternal relatives, paternal relatives, or other caregivers. If the new adult assumes a central role in household decision-making, the child’s relationship with the head of household or primary caregiver may change, either positively or negatively. When the new head of household has a low degree of kinship—or no kinship at all—with the orphan, their willingness to invest in the child’s education may be reduced. This dynamic is supported by Hamilton’s rule, which posits that an individual’s willingness to act altruistically toward another is determined by the degree of genetic relatedness between them. Thus, a caregiver with distant family ties would have fewer incentives to make educational investments in the child than a biological parent. Even if educational investment is maintained, the orphan’s schooling may be indirectly affected if the child is required to assume greater domestic responsibilities or is assigned other tasks that interfere with school attendance. This reallocation of roles within the household may reflect not only a lower priority assigned to the orphaned child but also a subtle form of discrimination within the household.

3.4 Expectations of Educational Returns

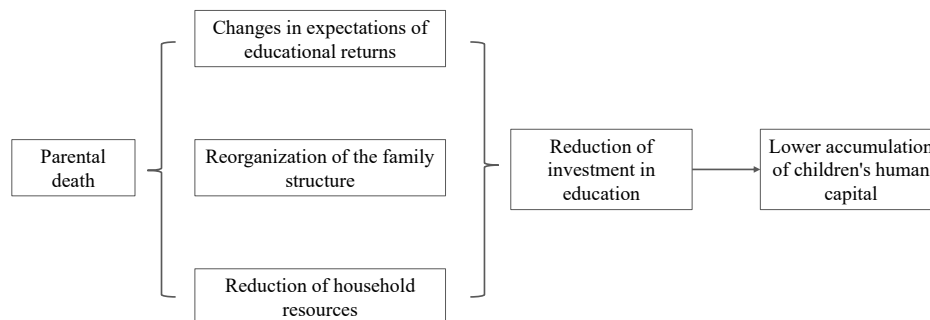
Following the loss of a parent, orphaned children often face a lack of attention, loss of emotional support, and adverse circumstances that affect their cognitive and socioemotional development (Case et al., 2004). These conditions can deteriorate their mental health,

limit their learning capacity, and reduce their academic performance, thereby reinforcing the household's perception that educational returns will be low.

This channel argues that households may make resource-allocation decisions based on the expected return from educational investment. If they perceive that the future benefits of schooling the orphaned child do not outweigh the costs, they may choose to reduce that investment and reallocate both resources and the child toward more profitable activities such as child labor or domestic work. This mechanism emphasizes the role of perceived opportunity cost: when education is no longer viewed as a worthwhile investment, the household adjusts its strategy according to what it considers most beneficial. Consequently, resources may be directed toward other household members with greater perceived potential, relegating the orphaned child to activities that generate immediate benefits for the household, even if this means abandoning or reducing their education.

In summary, these mechanisms are synthesized in Figure 1:

Figure 1: Transmission mechanism of parental death on the formation of child human capital



Note: Author's own work (2025).

4 Empirical Strategy

Parental death constitutes an exogenous event that justifies the use of quasi-experimental methodologies. Although some illnesses may allow for a certain degree of anticipation, death

is not a deliberate decision, which minimizes endogeneity problems that could bias the results. Therefore, a natural approach for this study is to employ a Difference-in-Differences (DiD) framework, which enables the identification of causal effects based on variations over time and across groups.

However, the classical Two-Way Fixed Effects (TWFE) model has important limitations when treatment is adopted in a staggered manner over time, as is the case here, since parental death occurs at different points for different individuals. Under a staggered treatment design, the TWFE estimator combines comparisons between groups treated at different times, which can generate biased estimates and even estimates with the wrong sign. For this reason, this study will use methodologies specifically designed for such contexts, such as the multi-period Difference-in-Differences approach proposed by Callaway and Sant’Anna (2021), which allows for a more coherent identification of effects in the presence of staggered treatments.

Additionally, the assumptions of the model are less restrictive than those of traditional Difference-in-Differences approaches. In particular, it allows for the inclusion of covariates that make the parallel trends assumption more plausible by controlling for factors that could influence the evolution of educational outcomes. This flexibility is especially valuable in educational contexts, where multiple factors interact and affect outcomes.

4.1 Identification Assumptions

The implementation of the staggered model requires the verification of three fundamental identification assumptions, which are analyzed below.

- a) **Irreversibility of treatment:** This assumption establishes that once a unit receives the treatment, it remains treated in subsequent periods. The death of a biological parent constitutes an irreversible event, thereby naturally satisfying this condition.
- b) **Limited treatment anticipation:** This assumption requires that the treatment cannot be anticipated by individuals, a condition that parental death satisfies, as it is an

unplanned event. However, even when this is met, individuals may modify their behavior in advance if they perceive the proximity of the event, as occurs with terminal illnesses. The model allows for the incorporation of this anticipation horizon through the parameter $\delta \geq 0$. In this study, we set $\delta = 0$ because the sample does not provide information to distinguish between sudden deaths and those preceded by periods of illness. This specification implies that treatment effects manifest starting from the period in which the death occurs.

c) Conditional parallel trends: The specification of this assumption depends on the choice of the control group. The “never-treated” group (children who never lost either parent) was chosen as the counterfactual. This choice is justified because, by controlling for appropriate covariates, systematic differences in the outcome between those who never experience the event and those who do can be captured. Subsequently, the “not-yet-treated” group (children whose parents are still alive but will eventually experience the death of one of them) will be used in robustness checks. Using the “never-treated” group as a control, the following condition must hold:

$$E[Y_t(0) - Y_{t-1}(0) \mid X, G_g = 1] = E[Y_t(0) - Y_{t-1}(0) \mid X, C = 1]$$

for all $g \in \mathcal{G}$ and $t \in \{2, \dots, T\}$ such that $t \geq g - \delta$, where X is a set of covariates. This assumption implies that, in the absence of treatment and conditional on certain covariates, the average evolution of the outcome for the cohort treated in year g would have been equal to the evolution of those who were never treated ($C = 1$). This assumption is not directly verifiable as it involves potential outcomes; however, empirical evidence supporting its plausibility is provided in the results section.

4.2 Model Specification

The specification differs from the traditional DiD by performing specific estimations by treatment cohort. Each cohort G_g comprises all individuals who experience parental death for the first time in period g . For each cohort, the average treatment effect on the treated (ATT) is estimated by comparing a post-treatment period t with a pre-treatment period $g - 1$. The parameter of interest is defined as:

$$ATT(g, t; \delta) = E \left[\left(\frac{G_g}{E[G_g]} - \frac{\frac{p_g(X)C}{1-p_g(X)}}{E \left[\frac{p_g(X)C}{1-p_g(X)} \right]} \right) (Y_t - Y_{g-\delta-1} - E[Y_t - Y_{g-\delta-1} | X, C = 1]) \right]$$

Where $\frac{p_g(X)}{1-p_g(X)}$ represents the inverse probability weights (IPW) and $E[Y_t - Y_{g-\delta-1} | X, C = 1]$ corresponds to the conditional prediction of the outcome regression.

The model uses doubly robust estimations to estimate specific ATTs for each combination of cohort and period, which are subsequently weighted to obtain a simple or dynamic aggregated effect. The first estimator is an outcome regression that predicts what would have happened to the treated units in the absence of treatment using observable characteristics and the dependent variable. The second component, the inverse probability weighting (IPW) estimator, uses the generalized propensity score to assign weights to the control units so that their distribution of covariates is comparable to that of the treated units. The doubly robust estimator combines both approaches, as it starts from the prediction of the outcome regression and adjusts it with the information contained in the IPW weights, reweighting the control group to make it equivalent to the treated group. Thus, the estimator remains consistent even if one of the two models is misspecified.

5 Data

5.1 Data and Sample Selection

The present work employs the Young Lives database, a longitudinal cohort study coordinated by the University of Oxford that has followed child trajectories in four low- and middle-income countries, Ethiopia, Vietnam, India, and Peru, over the last 20 years through seven survey rounds: 2002, 2006, 2009, 2013, 2016, 2020, and 2023 (Cueto and Espinoza, 2025). In each round, standardized information was collected on different dimensions of child and family well-being with the aim of generating reliable evidence that allows policymakers to design effective social programs oriented toward improving the living conditions of children and adolescents in situations of poverty.

In the case of Peru, to meet the central objectives of Young Lives, a sampling procedure was designed in which poor areas were over-represented and the top 5% of the richest districts in the country were deliberately excluded. Consequently, the sampling approach applied in Peru differs from a conventional randomized cluster design and from its origin was conceived as a study focused on childhood in contexts of vulnerability. To this end, in 2002, 20 districts distributed across 14 regions of the country were selected through a “sentinel site sampling” scheme that combined the deliberate choice of localities with socioeconomic diversity with a random selection of households within each locality. From this process, two cohorts were formed: a younger one and an older one, with an average difference of 7 years between each group.

For the present work, the analytical sample is restricted to the younger cohort, given that the interest is to evaluate the effects of orphanhood on education during childhood. The older cohort, which started the study at 7–8 years of age in 2002, reaches legal age by the fourth round, making it impossible to observe the childhood educational effects that constitute the focus of this analysis. In contrast, the younger cohort, comprised of 2,052 children born between 2001 and 2002, remains within the age range of childhood and adolescence

throughout the first 5 rounds, allowing the study of educational trajectories in the face of family shocks with a differences-in-differences model.

Delving into the previous paragraph, specific cleaning criteria per round are applied within the younger cohort. In round 1 (2002), those cases in which the death of one or both biological parents was already reported are eliminated, since the design requires children to begin follow-up with living parents. In round 2 (2006), parental death cases are also excluded, given that the baseline is established at this stage and it is necessary to have clean pre-event observations; additionally, children who die or drop out of the study in this round are eliminated, as they do not provide sufficient information for the analysis. In round 3 (2009), children who die or drop out are discarded due to the lack of minimum trajectories, but not those who experience parental loss, who remain in the sample as they have sufficient observation periods for comparison. From round 4 (2013) and in round 5 (2016), children who die or drop out are not retroactively eliminated, but are maintained in the sample until the last point where valid information is available. Finally, Rounds 6 (2020) and 7 (2023) are not included, given that by then the children of the younger cohort have reached legal age and, additionally, the effects of the pandemic affect the comparability of educational measurements.

As a result of these cleaning processes, the sample size of the younger cohort was progressively reduced between rounds, passing from 2,037 children in round 1 (2002) to 1,920 in round 2 (2006), 1,873 in round 3 (2009), 1,813 in round 4 (2013), and 1,739 in round 5 (2016). This phenomenon, known as attrition between rounds, reflects the cumulative loss of nearly 15% of the initial sample. The largest drop occurred between the first and second rounds, with a reduction of 117 children, equivalent to 5.7% of the sample, while in the following rounds the losses were smaller in both absolute numbers and proportion. The reasons behind these reductions include the death of some children, logistical difficulties in follow-up, and the decision of households not to continue participating.

In regards to the sample, to evaluate potential initial differences between children who

will not become orphans and those who are going to lose one of their parents, descriptive statistics and difference-of-means tests were calculated. For this analysis, Welch's t-test was used instead of the standard t-test that assumes equal variances between groups. This choice is based on the fact that Welch's t-test provides better control of Type I errors when the assumption of homogeneity of variances is not met and maintains a robustness comparable to Student's t-test when the assumptions are met. In particular, the assumption of homogeneity of variances is less feasible when sample sizes are very different, as occurs in this study, because the variance of the small group is expected to be larger and this could distort the results of the standard t-test.

The variables considered in this analysis encompass individual characteristics of the child, household composition, and household conditions. Among the individual characteristics of the child, gender (male dummy) is included, as well as their percentage of correct answers in the Peabody Picture Vocabulary Test (PPVT), a standardized test that measures receptive vocabulary through the choice of images that represent heard words, and whether they suffer from a chronic (permanent) or acute (temporary) illness. Regarding household composition, the age of the household head, the parents' Spanish fluency, and the number of household inhabitants are included. Regarding household conditions, the household wealth index, the existence of debts, access to basic services (drinking water, sanitation, electricity), and access to social programs are included. Finally, among external factors, crop loss is contemplated, as well as rural residence status. Below, the difference-of-means table (Welch's t-test) for these characteristics is presented:

The results of Welch's test indicate that there are no statistically significant differences between children who will never become orphans and those who will lose one of their parents in the future. Individual, household, and environmental characteristics show very similar averages between both groups, with p - values above significance thresholds. This absence of statistically relevant differences suggests that the groups are balanced on observable variables, which supports the validity of the comparisons presented in subsequent analyses and

Table 1: Descriptive statistics and difference-of-means between groups according to orphanhood status

Variable	Total Sample	Never Orphans	Future Orphans	Diff. Means	P-value
Gender (male=1)	0.50	0.50	0.49	0.01	0.85
Peabody Picture Vocabulary Test	0.64	0.65	0.62	0.03	0.22
Chronic Illness (suffers=1)	0.10	0.10	0.07	0.03	0.48
Acute Illness (suffers=1)	0.20	0.21	0.16	0.04	0.42
Father speaks Spanish (speaks=1)	0.99	0.99	0.98	0.01	0.64
Mother speaks Spanish (speaks=1)	0.92	0.92	0.89	0.03	0.47
Age of household head	38.34	38.30	39.96	-1.67	0.23
Household size	5.51	5.50	5.73	-0.22	0.44
Rural (rural locality=1)	0.30	0.30	0.36	-0.06	0.35
Wealth index	0.47	0.47	0.43	0.05	0.18
Household with debts (debtor=1)	0.25	0.25	0.22	0.03	0.60
Drinking water (access=1)	0.50	0.50	0.51	-0.01	0.89
Sanitation (access=1)	0.81	0.81	0.87	-0.06	0.19
Social programs (access=1)	0.69	0.69	0.75	-0.05	0.37
Electricity (access=1)	0.82	0.82	0.80	0.02	0.77
Crop loss (loss=1)	0.03	0.03	0.02	0.01	0.64
N	1920	1865	55	1920	1920

reinforces that this round constitutes a valid baseline for the study.

5.2 Variables

The central independent variable of this study is parental death, defined as the loss of at least one of the biological parents during the observation period. This is coded as a dichotomous variable that takes the value 1 starting from the round in which the death of the biological father or mother is reported for the first time, and remains active in subsequent rounds. In case of multiple losses, meaning the death of both parents at different times, the first reported event is considered as the main treatment.

The dependent variables of the study focus on different dimensions of child education: (1) annual household educational expenditure, constructed as the sum of the amounts reported in the last 12 months for enrollment, tuition, school supplies, transportation to the educational institution, school fees, and private tutoring; (2) school enrollment, a dichotomous variable

that captures whether the child regularly attends their educational institution; (3) hours at school, which measure the daily time dedicated to attending classes at the educational institution; and (4) study hours at home, which collect the daily time invested in educational activities outside of school.

Likewise, the analysis incorporates three additional variables: hours dedicated daily to domestic chores, hours in unpaid work, and hours in paid work. These do not represent dimensions of education, but allow observing how orphanhood may affect the child’s time distribution. For example, an increase in these activities, accompanied by a reduction in time allocated to leisure or education, could reflect a displacement of temporal resources that helps to better understand the mechanisms through which parental loss affects child development.

Finally, the household wealth index and the number of household inhabitants are incorporated as control variables. The wealth index, constructed directly by Young Lives, combines information on different dimensions of the material well-being of the household and is structured into three equally weighted sub-indices: (i) housing quality, considering materials of floors, roofs, and walls, number of rooms, and crowding (household size divided by the number of rooms); (ii) access to basic services, such as drinking water, sanitation, electricity, and cooking fuel; and (iii) possession of durable goods, which for the case of Peru includes radio, television, bicycle, motorcycle, automobile, landline telephone, mobile telephone, refrigerator, stove, blender, iron, and record player. The values of the three sub-indices are averaged to obtain the final wealth index, where a higher value indicates a higher socioeconomic status (Briones, 2017).

For its part, the number of people living in the household shows how the family is formed and the economic pressures that can arise regarding how much is invested in education. Households with more members usually have fewer resources available per person, which can reduce educational expenditure. In contrast, smaller households usually have more resources per member and can invest more in the education of their children. Controlling for this

variable is important to better identify the real effect of the death of one of the parents on educational investment, without household size confounding the results. The use of these controls is essential in the context of the study, as it allows for a better distinction of the impact attributable to parental death from the effects that stem from the demographic structure of the household, socioeconomic conditions, and relative availability of resources.

6 Results

The model is used to estimate effects on all the dependent variables detailed previously, both at the aggregate level and broken down by time relative to the event. Household wealth index and household size are included as controls, and standard errors are clustered at the sentinel site level. Table 2 and Figures 2 - 5 present the average effects by time relative to treatment. These results allow for the evaluation of the parallel trends assumption and the observation of the evolution of the effects after parental loss.

Table 2: Dynamic Aggregate ATT

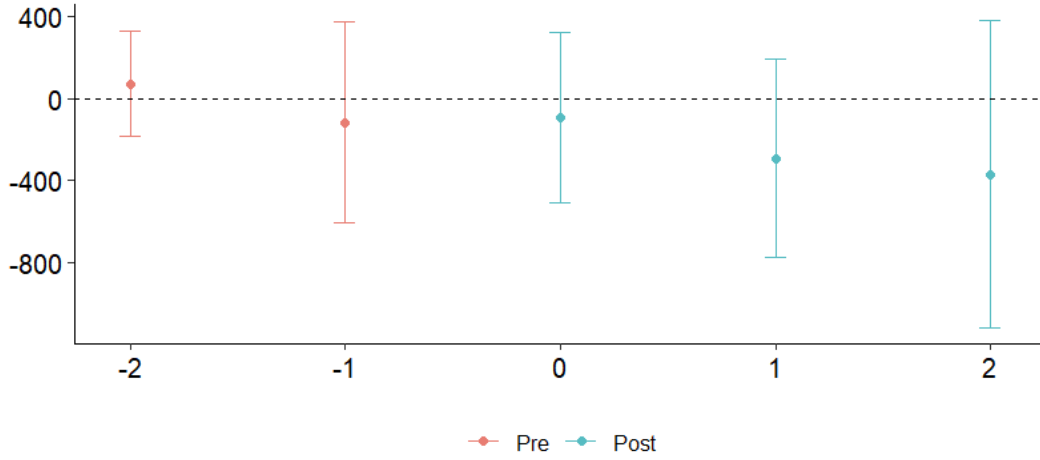
Dependent variable	-2	-1	0	1	2
Education spending	70.8789	-119.5337	-95.2893	-293.0605	-370.1609
Enrollment	0.0907	-0.0394	-0.0905*	-0.1912	-0.1416*
School hours	0.3752	-1.1883	-0.6242	-1.1089	-0.8324
Study hours	0.1922	0.0741	-0.4935*	0.0573	-0.6204
Housework hours	0.2225	-0.2736	0.0071	-0.2719	-0.4073
Paid work hours	-0.0042	-0.0433*	0.3218	-0.0960*	0.6699
Unpaid work hours	0.0819	0.1515	0.2763	0.2599	-0.2508

Significance codes: * the interval does not cover the value zero at 95%

As seen in Figure 2, the pre-treatment coefficients ($t = -2$ and $t = -1$) are not significant, which supports the parallel trends assumption. Subsequently, a progressive and persistent drop in education spending measured in real 2021 terms is observed: -293.1 soles at $t = 1$ and -370.2 soles at $t = 2$, although neither of these effects is statistically significant at 95%. This suggests a downward trend in household educational investment after parental loss, but without conclusive statistical evidence of an abrupt change. In the period

when the event occurs ($t = 0$), the effect is also negative (-95.3 soles) and not significant, which would indicate that households initially maintain their education spending but show a gradual adjustment in the following periods, possibly associated with income reduction or the reorganization of the family structure.

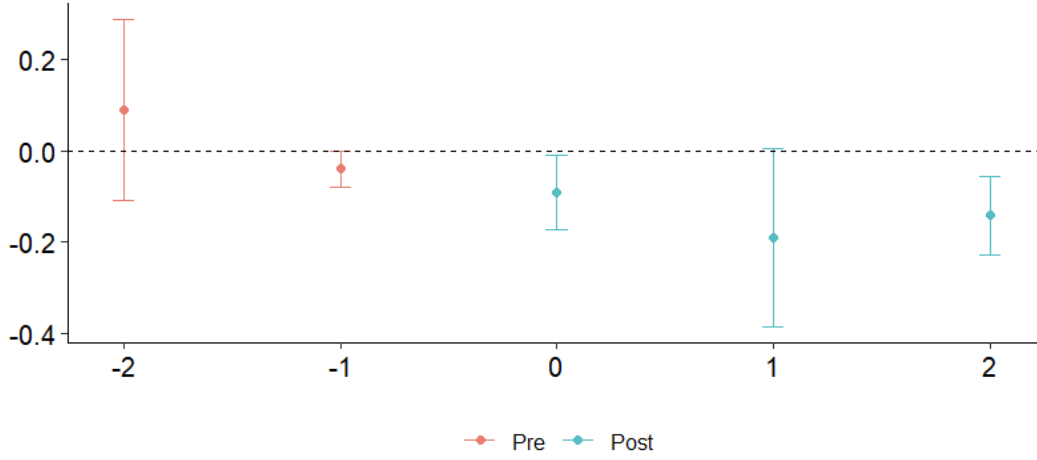
Figure 2: Dynamic aggregate ATT for education spending



As shown in Figure 3, the pre-treatment coefficients ($t = -2$ and $t = -1$) are small and not significant, supporting the parallel trends assumption. Following parental loss, the effects become negative and statistically significant in the event period ($t = 0$), with a drop of 0.09 percentage points in the probability of being enrolled. Subsequently, at $t = 1$, enrollment decreases by 0.19 percentage points, although this effect is not significant. At $t = 2$, a negative and significant effect of -0.14 percentage points is observed again, suggesting that parental loss has an immediate and persistent impact on children's school continuity.

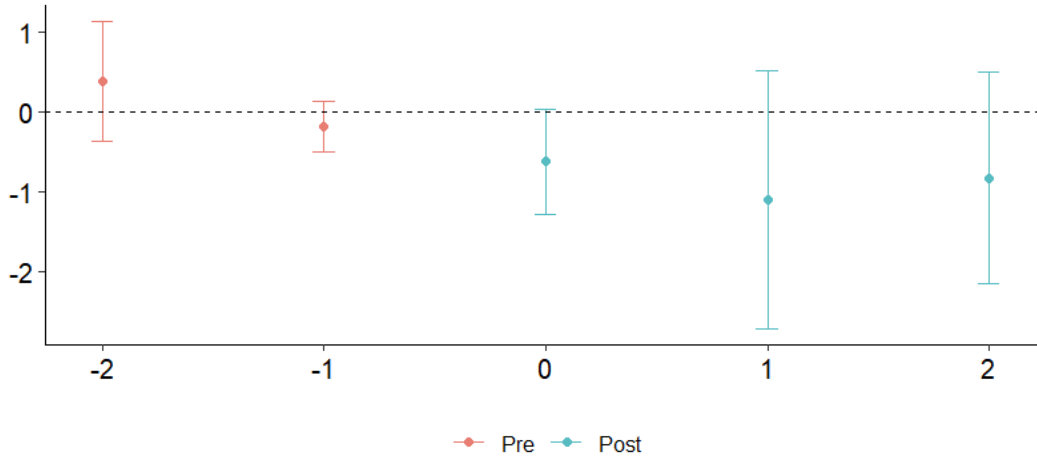
In Figure 4, the pre-treatment coefficients are small and not significant, confirming the absence of systematic differences between orphans and non-orphans before the event. After parental loss, the effects turn negative in all subsequent periods, but none reach statistical significance at 95%. At $t = 0$, the reduction is -0.62 hours (approx. -37 minutes), widens to -1.11 hours at $t = 1$ (approx. -67 minutes) and ends at -0.83 hours at $t = 2$ (approx. -50 minutes). Taken together, these results indicate a sustained decrease in daily time spent at school after parental loss, although statistical evidence does not allow confirming that this

Figure 3: Dynamic aggregate ATT for enrollment



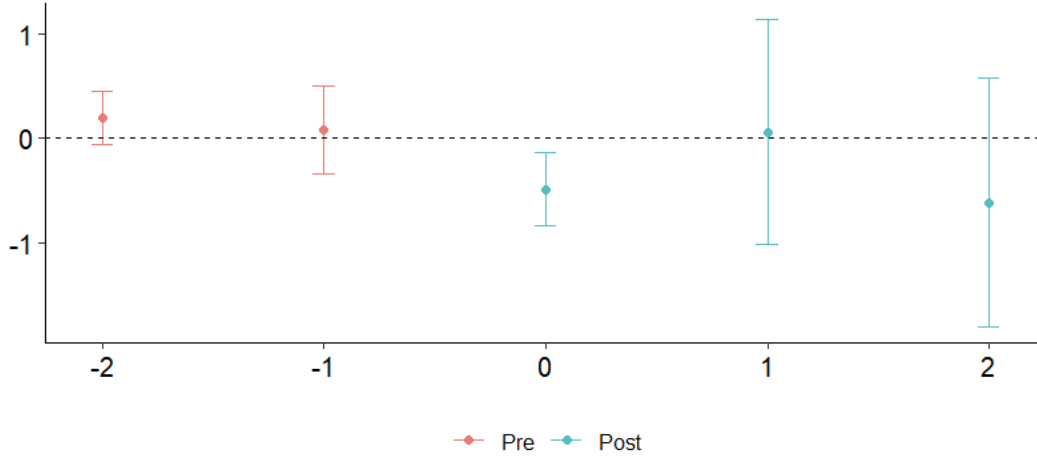
drop is significant.

Figure 4: Dynamic aggregate ATT for school hours



In Figure 5, the pre-treatment coefficients are small and not significant, supporting the parallel trends assumption. In the event period ($t = 0$), there is a significant reduction of -0.49 hours (approx. -29 minutes) in time spent studying outside of school. At $t = 1$, the effect changes sign and becomes slightly positive (0.06 hours, approx. 4 minutes), while at $t = 2$ it becomes negative again (-0.62 hours, approx. -37 minutes), though without statistical significance. These results indicate that parental loss generates an immediate and statistically significant adjustment in study routines.

Figure 5: Dynamic aggregate ATT for study hours



Regarding the other dimensions of time use, the results suggest an absence of systematic effects associated with parental loss. In the case of housework, the coefficients are small and not significant across all periods, indicating stability in this activity before and after the event. In contrast, paid work shows significant effects at $t = -1$ (-0.04 hours, approx. -2 minutes) and at $t = 1$ (-0.10 hours, approx. -6 minutes); both are small in magnitude but suggest minor adjustments in time allocation toward productive activities around the time of the event. On its part, unpaid work does not present significant effects, despite registering positive fluctuations before and after the loss. Together, these patterns confirm that parental loss does not generate a substantial reallocation of time toward domestic or productive work, and that its main consequences are concentrated in the immediate reduction of enrollment, study time, and, to a lesser extent, a downward trend in education spending.

Regarding the aggregate effect, the results presented in Table 3 summarize the average effect of parental loss on each variable during the post-event window. It is worth noting that it was not weighted by cohort size, as this can be biased toward early-treated individuals, who have more post-treatment observations. Instead, it was weighted by exposure time, which allows for the consideration that treatment does not occur homogeneously across different individuals.

First, an average reduction of 252.8 soles in household education spending is observed,

although this effect is not statistically significant at 95%. While the negative sign is consistent with a decrease in educational investment after the parental death, the magnitude and lack of significance indicate that the evidence of a sustained impact on education spending is limited.

In the case of school enrollment, the estimated ATT is -0.1411 , significant at 95%, implying that orphaned children have, on average, a 14.1 percentage point lower probability of being enrolled compared to non-orphans. This result confirms that parental loss has a direct and significant effect on school continuity.

Similarly, school attendance hours show an average reduction of 0.86 hours per day (approximately 51 minutes), which is significant at 95%, suggesting a substantive and sustained decrease in the actual time children spend at school after the parental death. On the other hand, time dedicated to studying outside of school presents an ATT of -0.35 hours (around 21 minutes), though this effect is not significant. The direction of the coefficient remains consistent with less dedication to studying outside of class after parental loss, but without sufficient statistical evidence to claim a change in this dimension.

In contrast, non-educational activities, such as housework hours, paid work, and unpaid work, do not present statistically significant effects. The coefficients are small and close to zero, reinforcing the idea that parental loss mainly affects areas directly linked to the educational process, without inducing a reallocation of time toward child labor or housework.

Table 3: Aggregate ATT

Dependent variable	ATT	Standard Error	95% Confidence Interval	
Education spending	-252.8369	230.2762	-704.17	198.4961
Enrollment	-0.1411*	0.0441	-0.2276	-0.0546
School hours	-0.8552*	0.3666	-1.5736	-0.1367
Study hours	-0.3522	0.2193	-0.782	0.0776
Housework hours	-0.224	0.2046	-0.6251	0.1771
Paid work hours	0.2986	0.3435	-0.3746	0.9718
Unpaid work hours	-0.0891	0.7223	-1.5048	1.3267

Significance codes: * the interval does not cover the value zero at 95%

Next, we analyze whether the primary channel through which parental death affects

educational outcomes is the household wealth channel. To do this, household education spending is considered the variable of interest. The hypothesis is that the drop in education spending, in percentage terms, should be larger among lower-wealth households, given that they are more vulnerable to this type of shock. Conversely, obtaining results that differ from this pattern would suggest that the wealth channel is not the main mechanism.

With this goal, the sample of treated children was divided into two groups: those above the median of the wealth index and those below the median (see Table 4). Subsequently, the model was estimated for each group separately to obtain the simple aggregate ATT—meaning weighted by cohort size and not by exposure time—since the objective here is not to capture the evolution of the effect over time, but to compare average magnitudes between wealth subgroups. The effect is expressed as the percentage drop in education spending relative to the average baseline spending before the event, calculated as the average of individual means. Ideally, the effect would have been compared in relation to household income; however, this variable is not available in the dataset for all rounds. The estimations control only for household size. Since the sample was already divided by wealth level, it would not be appropriate to control for this variable again. Furthermore, while other variables could influence education spending trajectories, the small size of the groups limits the possibility of including more controls without affecting the precision of the estimations.

The results (Table 5) show that, although the drop is larger in the lower-wealth group, the difference between the two groups is not substantial. Combined with the previous results, which indicated that the reduction in education spending was not significant in the immediate period following the event but rather later on, these findings suggest that while the wealth channel is relevant, it does not seem to be the primary mechanism through which parental death affects educational outcomes.

Now we will analyze the educational returns channel. This mechanism suggests that parental death can psychologically affect children, impacting their socioemotional and cognitive development as well as their learning capacity. Faced with this situation, families

Table 4: Distribution of Treated Individuals Based on the 50th Percentile of the Wealth Index

Group	Individuals	Observations
Above 50th percentile	27	122
Below 50th percentile	28	96

Table 5: Results by Wealth Index Subgroup

Variable	ATT	Standard Error	Baseline Average	Drop
Education spending (upper group)	-503.3724	316.2924	935.6821	53.80%
Education spending (lower group)	-146.7746	82.5148	225.9931	64.95%

Significance codes: * the interval does not cover the value zero at 95%.

re-evaluate whether education remains a profitable investment after the shock. For children with good prior academic performance, households are expected to trust that they will recover from the shock and thus maintain their educational investment. Conversely, if the child shows low performance, the family might consider it more beneficial to reallocate their time toward economic or domestic activities that generate immediate benefits.

Since we do not have grades or cognitive test scores for all periods, we use the PPVT test score as an indicator of prior performance. Using this indicator, we split the children into two groups: those with scores above the 75th percentile and those below it (see Table 6). In this case, the variable of interest is school enrollment, and we expect to observe that for the higher-performance group, enrollment decreases little after the parental death; whereas, for the lower-performance group, enrollment drops more sharply and the time spent on other activities increases. The results (Table 7) partially confirm this hypothesis because the drop in enrollment for the lower group is twice that of the upper group. However, the differences in hours spent on work and housework are not statistically significant.

Table 6: Distribution of Treated Individuals Based on the 75th Percentile of the PPVT Test

Group	Individuals	Observations
Above 75th percentile	16	119
Below 75th percentile	39	99

Table 7: Results by PPVT Test Subgroup

Variable	ATT	Standard Error
Upper group		
Enrollment	-0.069*	0.0326
Housework hours	-0.1284	0.2423
Paid work hours	0.053	0.2005
Unpaid work hours	-0.0284	0.3611
Lower group		
Enrollment	-0.1309*	0.0459
Housework hours	0.0094	0.2406
Paid work hours	0.3727	0.3564
Unpaid work hours	-0.1972	0.6324

Significance codes: * the interval does not cover the value zero at 95%

7 Robustness Checks

In order to evaluate the sensitivity of the results, an alternative specification was estimated by modifying the set of controls included in the model and the comparison group. In the main estimation, household wealth index and household size were controlled for, as these variables capture baseline differences in financial resources and family composition that can lead to different educational trajectories. However, given the limited size of the treated cohorts, incorporating a high number of controls in the baseline specification was avoided to protect the precision of the estimations.

In this alternative version, the "not yet treated" individuals—those who have not yet received the treatment but will in later periods—were added to the comparison group. Likewise, the set of controls from the baseline model was replaced by variables that capture equivalent dimensions but through a more parsimonious specification. Instead of the wealth index and household size, a rural zone dummy was included to reflect structural differences in the availability and quality of educational opportunities, along with the number of siblings under 18, which captures the internal demographic pressure of the household on educational time and resources. This modification allows for the evaluation of the sensitivity of the results against a different set of controls that maintain the same conceptual logic, without

over-fitting the model.

The results are presented in Table 8. In general, the coefficients maintain the direction observed in the main estimation, though with greater magnitude and statistical significance, suggesting high robustness of the estimated effect. Education spending now shows an average reduction of 534.2 soles, significant at 95%, reinforcing the evidence of a substantial drop in household educational investment after parental loss. Consistently, the probability of enrollment remains negative and significant, with an effect of -0.16 percentage points, very similar to that of the baseline model.

Likewise, school attendance hours retain a negative and significant effect of -1.05 daily hours (approx. 63 minutes less), slightly higher than estimated in the main specification (-0.86 hours). This result suggests that parental loss systematically reduces children's time in the school environment, even after controlling for regional differences and family composition.

An important difference from the main model is that, in this version, study time outside of school also becomes significant at 95%, with a reduction of 0.50 daily hours (approx. 30 minutes). This reinforces the hypothesis that parental loss limits not only school attendance but also complementary academic effort outside the classroom.

On the other hand, non-educational variables (housework, paid work, and unpaid work) continue to show small and non-significant coefficients, in line with the baseline model. This indicates that parental loss does not induce a reallocation of time toward productive or domestic activities, and that its effects are robustly concentrated in the dimensions directly linked to the educational process.

A limitation of linear difference-in-differences models is that they impose a linear relationship between the outcome variable and the treatment, without considering the nature of the outcome. In this paper, the variables of interest, such as school enrollment (a binary variable), household education spending (a positive continuous variable), and study hours (a count variable), are not strictly continuous, meaning the linear specification may not adequately reflect their distribution.

Table 8: Aggregate ATT with New Specification

Dependent variable	ATT	Standard Error	95% Confidence Interval	
Education spending	-534.218*	202.842	-931.781	-136.655
Enrollment	-0.1631*	0.0362	-0.2344	-0.0927
School hours	-1.0482*	0.265	-1.5676	-0.5289
Study hours	-0.5005*	0.2438	-0.9784	-0.0225
Housework hours	-0.0455	0.1851	-0.4084	0.3174
Paid work hours	0.1845	0.7908	-1.3655	1.7344
Unpaid work hours	0.2828	0.2094	-0.1275	0.6932

Significance codes: * the interval does not cover the value zero at 95%

To evaluate the robustness of the results, the methodology proposed by Wooldridge (2023) is used, which develops a simple extension of the difference-in-differences method for non-linear models with panel data. This approach is based on assuming a functional form compatible with the nature of the dependent variable within the exponential family. For school enrollment, which is a dichotomous variable, a logistic function is used, appropriate for modeling probabilities between 0 and 1. For household education spending, which is a continuous variable greater than or equal to zero, a Poisson model is estimated by maximum likelihood, which guarantees non-negative predictions. For study hours, which constitute a count variable, a Poisson model is also applied, as it is appropriate for discrete and non-negative data.

The results presented in Table 9 generally confirm the direction of the effects observed in the linear estimations, though with some differences in magnitude and significance. First, household education spending presents a negative effect of -183.3 soles, without statistical significance at 95%. Although the magnitude is smaller than in the main linear model (-252.8 soles) and the robustness estimation (-534.2 soles), the sign remains consistent, suggesting that parental loss tends to reduce education spending, albeit with weaker evidence under the non-linear specification.

In the case of school enrollment, the estimated effect is also negative (-0.049) and not significant, in contrast to the linear models where the impact was significant and close to -0.14 percentage points. On the other hand, school attendance hours show a major change

compared to the main model, as the estimated effect is positive and statistically significant at 0.1%, with an average increase of 0.98 daily hours. Unlike the linear models, which yielded significant reductions in attendance (-0.86 to -1.05 hours), this result suggests that when modeling the variable within an appropriate distribution, parental loss could be associated with longer time spent at school, which might be explained by heterogeneous responses.

Finally, study time outside of school presents a slight and non-significant effect, consistent with previous results that also did not show an impact statistically different from zero. The non-linear estimations confirm the general direction of the linear results for most variables, though they reveal a clear difference in magnitude and significance regarding school attendance.

Table 9: Marginal Effects

Dependent variable	Marginal Effect	Standard Error	P-value
Education spending	-183.2683	115.8785	0.114
Enrollment	-0.0488394	0.0427328	0.253
School hours	0.9780263***	0.3080413	0.001
Study hours	-0.0249238	0.1444865	0.863

Note: * $P \leq 0.05$; ** $P \leq 0.01$; *** $P \leq 0.001$

8 Limitations

The main limitations of the study stem from the nature and low frequency of the event analyzed. First, the rarity of parental death during childhood means there is a small number of treated observations, which can affect the statistical precision of the estimations and limit the power of significance tests. Second, the possibility of anticipated behavior cannot be completely ruled out, especially when the death occurs after a long illness. In such cases, families might change their decisions regarding spending or school attendance before the actual death happens, affecting the validity of the causal interpretation. Finally, the panel structure has important limitations, as there are missing values, variations in how variables are measured, and a loss of information between rounds. This restricts the inclusion of

certain controls and forces the model to keep a parsimonious specification. This lack of harmonization may also explain part of the instability in the results between linear and non-linear models.

9 Conclusion

This study analyzed the impact of losing a parent during childhood on children’s education in Peru, using longitudinal data from the Young Lives study and a staggered difference-in-differences methodology. The results confirm that the loss of a parent has negative and lasting effects on different areas of the children’s educational process.

In the linear estimations, losing a parent is linked to significant cuts in household education spending and enrollment probability, as well as a steady decrease in school attendance hours. However, when using non-linear models that adjust the functional form to the nature of each variable, the negative effects lose statistical significance, and in the case of school hours, the sign reverses. This contrast suggests that the relationship between orphanhood and education might be heterogeneous, depending on how the family adjusts and the specific limitations of each household after losing a parent. Likewise, the results show that non-educational activities—such as paid work, unpaid work, or housework—do not show significant effects. This supports the idea that losing a parent mainly affects educational areas rather than how time is allocated to productive activities.

In the analysis of transmission channels, estimations by wealth level show that the drop in education spending is more severe among relatively poor households, though negative effects are also seen in wealthier households. On the other hand, the analysis of the educational returns channel indicates that families might adjust their investment based on the child’s past performance; children with higher test scores tend to stay enrolled, while those with lower performance experience sharper declines. However, since the results are not conclusive, it is not possible to identify with certainty which channel is the main one. This leaves open

the possibility that the most relevant mechanism is the reorganization of the household after losing a parent—an aspect that is not covered in this study and remains open for future research.

From a public policy perspective, the results of this study highlight the importance of strengthening social protection programs aimed at orphaned children and teenagers. The effectiveness of these policies will depend on their ability to address the different channels through which losing a parent affects education: the reduction of household financial resources, family reorganization, and expectations regarding educational returns. In particular, policies that ensure school continuity and provide comprehensive support to the child after the loss can be crucial to prevent this event from turning into long-term educational delays and, ultimately, into lower human capital accumulation over their lifetime.

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